

Online Retail Datamining Final Report

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Abstract

This report presents the findings of a retail data mining project on the UCI Online Retail dataset. Three discovery questions motivate three analyses: association rule mining via Apriori, customer segmentation via RFM K-Means clustering, and seasonal pattern discovery via STL decomposition and DTW clustering. Key findings include strong market basket rules with lifts of up to 24, a null result in customer segmentation indicating a homogeneous customer base, and three distinct seasonal product clusters.

1 Introduction

1.1 Dataset

This project utilizes the **Online Retail Dataset** from the UCI Machine Learning Repository, containing invoice records from a UK-based online retailer. The retailer in question mostly sells knick-knacks and consumer items such as teacups, garden kneeling pads, alarm clocks, and gift items.
Data Source: <https://archive.ics.uci.edu/dataset/352/online+retail>

1.1.1 Data Characteristics.

- **Raw size:** 541,909 rows, 8 columns.
- **After cleaning:** 530,104 rows (97.9% retained).
- **Coverage:** 1 December 2010 – 9 December 2011 (one year).
- **Scope:** 4,338 unique customers, 3,922 unique products, 19,960 invoices.
- **Geography:** 91.5% UK transactions; secondary markets include Germany, France, and Ireland.

1.1.2 Schema. The primary attributes utilized in this analysis are detailed in Table 1.

Table 1: Dataset Schema

Feature Name	Type	Description
InvoiceNo	String	6-digit unique ID. ‘c’ prefix indicates cancellation.
StockCode	String	5-digit unique product ID.
Description	String	Nominal product name.
Quantity	Integer	Units per transaction.
InvoiceDate	Date	Timestamp of transaction.
UnitPrice	Float	Product price per unit (GBP).
CustomerID	Integer	5-digit unique customer ID.
Country	String	Name of the customer’s country.

1.2 Discovery Questions

- (1) What products are frequently purchased together?

- *Value:* Product co-purchase patterns enable cross-sell recommendations, bundle promotions, and grouped inventory management.
 - *Technique:* Market Basket Analysis (Apriori)
- (2) What is the distribution of spending across the customer base?
- *Value:* Identifying distinct customer segments allows tailored marketing strategies and prioritized re-engagement efforts.
 - *Technique:* Customer Segmentation (RFM K-Means Clustering)
- (3) Which product categories share similar seasonal sales patterns?
- *Value:* Seasonal clusters inform inventory stocking schedules and promotional timing.
 - *Technique:* Time Series Clustering (STL + Dynamic Time Warping)

1.3 Project Goals

- Apply three complementary data mining techniques to a real-world retail dataset.
- Follow the Knowledge Discovery in Databases (KDD) pipeline from raw data ingestion through to business insights.
- Evaluate the validity and limitations of each result.

2 Methods

2.1 Preprocessing

All notebooks share a common cleaned dataset produced by 01_preprocessing.ip

- Removed cancelled invoices (InvoiceNo prefixed with ‘c’).
- Removed rows with non-positive Quantity or UnitPrice.
- Stripped whitespace from product descriptions.
- Derived TotalPrice = Quantity × UnitPrice.
- Rows missing CustomerID retained for market basket analysis but excluded from the RFM matrix.

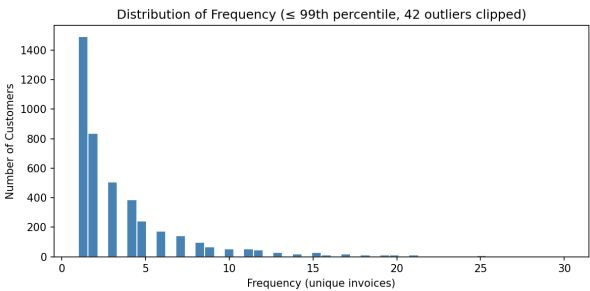


Figure 1: RFM Feature Distributions

2.2 Market Basket Analysis — Apriori

- **Data transformation:** Transactions encoded as a binary invoice $\times$ item matrix.
- **Algorithm:** Apriori. min_support swept across 7 thresholds (0.01–0.05); final value 0.02 selected.
- **Rule filtering:** Confidence ≥ 0.5 and Lift ≥ 2 .

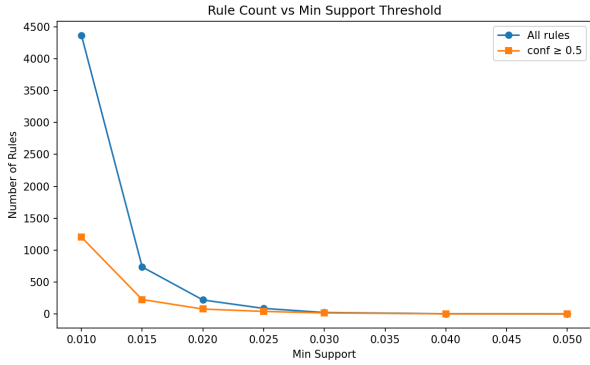


Figure 2: Apriori Hyperparameter Tuning

The overall Apriori pipeline is illustrated in Figure 3.

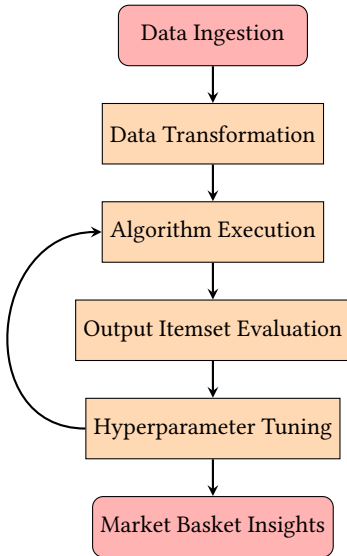


Figure 3: Apriori Analysis Pipeline

2.3 Customer Segmentation — RFM K-Means

- **Feature engineering:** Recency (days since last purchase), Frequency (unique invoice count), and Monetary (total spend) computed per customer for 4,338 customers with valid Customer ID.
- **Preprocessing:** log1p transform applied to all three features to reduce outlier influence + StandardScaler normalization (mean = 0, std = 1).

- **Algorithm:** K-Means (n_init=10, random_state=42) tested for $k \in [2, 10]$.
- **Model selection:** Elbow method and silhouette score; $k = 2$ selected.

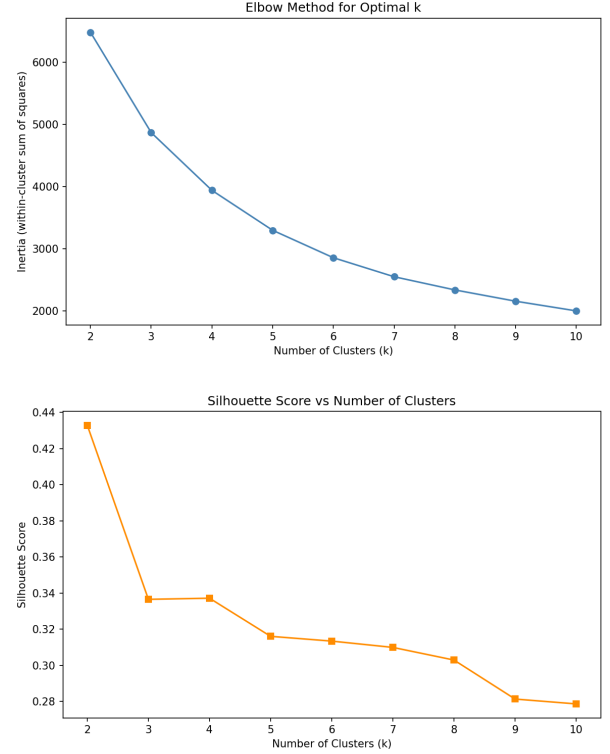


Figure 4: K-Means Cluster Count Selection: Elbow Plot (top) and Silhouette Score (bottom)

2.4 Time Series Clustering — STL + DTW

- **Category assignment:** 15 product categories assigned arbitrarily via keyword regex matching on product descriptions.
- **Aggregation:** Daily sales summed per category, then resampled to weekly totals.
- **STL decomposition:** Each weekly series decomposed into trend, seasonal, and residual components. Only the seasonal component is passed to clustering.
- **Algorithm:** TimeSeriesKMeans with DTW metric, $k \in [2, 7]$; $k = 3$ selected).

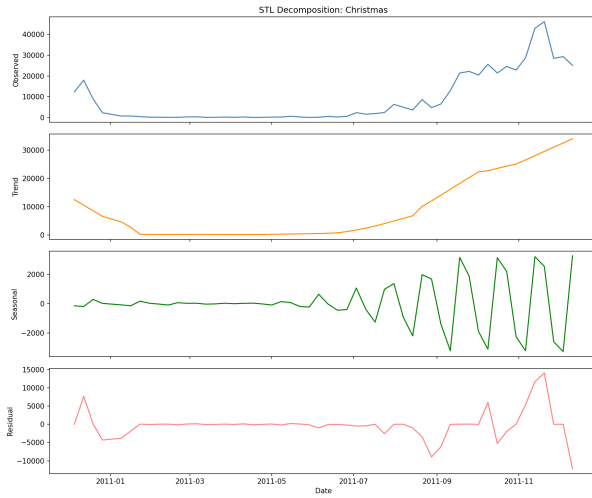


Figure 5: STL Decomposition – Christmas Category

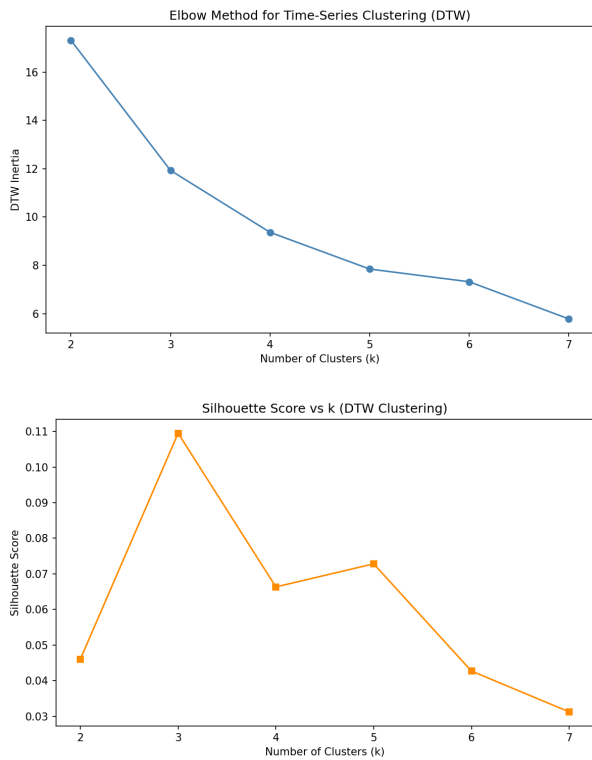


Figure 6: DTW Cluster Count Selection: Elbow Plot (top) and Silhouette Score (bottom)

3 Results

3.1 Market Basket Analysis

- 401 frequent itemsets identified at $\text{min_support} = 0.02$.
- 218 total association rules generated with 30 strong rules.

- **Dominant pattern:** Customers often purchase entire color-variant sets of a product rather than individual items.
- **Top rule:** {Pink Regency Teacup, Roses Regency Teacup} $\Rightarrow$ {Green Regency Teacup}; Support: 0.020, Confidence: 0.89, Lift: 24.2.

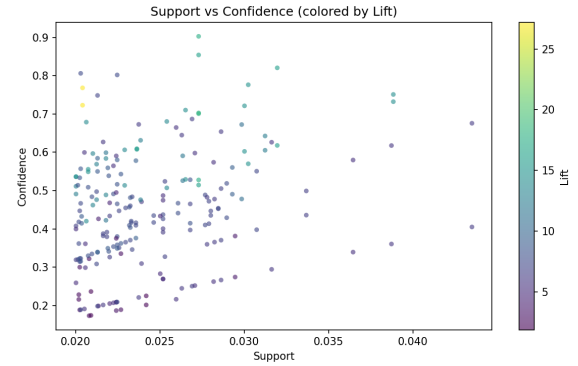


Figure 7: Association Rules: Support vs. Confidence (color = Lift)

3.2 Customer Segmentation

- RFM analysis did not reveal meaningful customer segments. Silhouette scores were low across all tested values of k , and elbow analysis showed no clear inflection point, indicating an absence of natural cluster structure in the data.
- FM values exhibit a tight linear correlation across the entire customer base. Customers are arranged on a homogenous spectrum of high to low spenders, not in any identifiable clusters.
- This constitutes a null result, which is itself informative. It suggests that spend-based marketing will be the superior approach as opposed to any attempt to target customer segments.

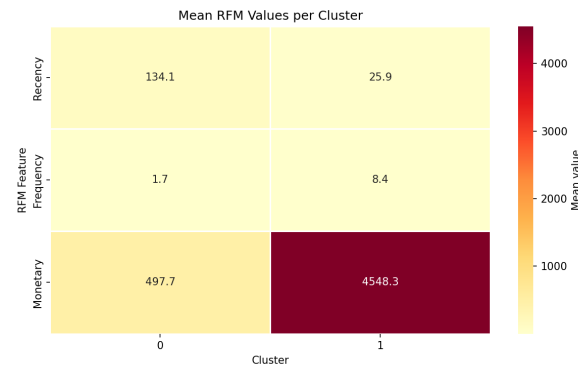


Figure 8: Mean RFM Values per Cluster

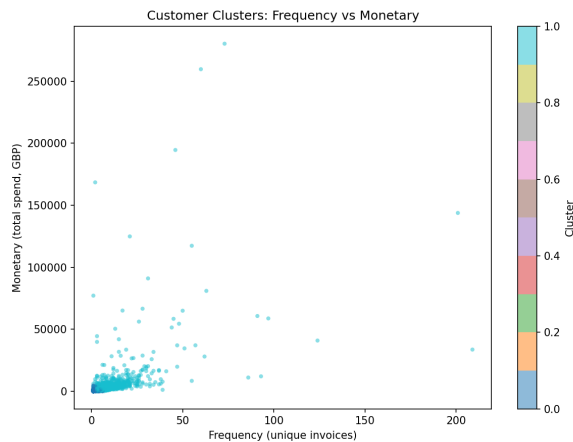


Figure 9: Frequency vs. Monetary by Cluster

3.3 Time Series Clustering

- $k = 3$ selected with DTW silhouette score of 0.11.
- **Cluster 0** (4 categories): Sharp Q4 holiday spike.
- **Cluster 1** (5 categories): Moderate Q4 peak with a trailing Q1 edge.
- **Cluster 2** (6 categories): Minimal seasonality and flat demand throughout the year.

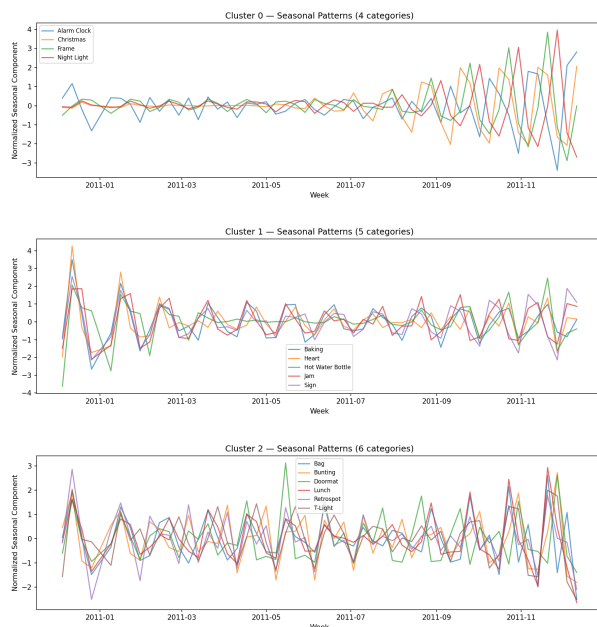


Figure 10: Seasonal Patterns per DTW Cluster: Christmas (top), Winter-Spring (middle), No Seasonality (bottom)

4 Interpretation

4.1 Market Basket Analysis

- The high-confidence, high-lift rules reflect deliberate collecting behavior from customers who seek to purchase multiple items in a variant set.
- At 89% confidence, the top rule in particular is near-universal among customers who initiate the behavior.
- *Business insight:* Cross-sell recommendations and inventory grouping should be structured around variant sets. Bundle promotions structured around variant set collection are likely to get high conversion rates.

4.2 Customer Segmentation

- The absence of multiple clusters in the FM space indicates that this retailer's customer base does not divide into distinct behavioral types. Customers vary in how much they spend, but not in how they spend.
- This is a common outcome in retail RFM analysis. The null result is informative and indicates that there is no reason to attempt to target customer behavior clusters that don't exist.
- *Business insight:* Simple spend-threshold targeting is likely to be the most effective strategy.

4.3 Time Series Clustering

- Cluster 0 products peak sharply and predictably in Q4, requiring pre-season stocking by Q3 and concentrated holiday promotional spend.
- Cluster 1 products have a broader winter relevance extending into Q1, suited to a longer promotional window spanning late autumn through early spring.
- Cluster 2 products carry flat year-round demand and are best managed on a lean continuous replenishment cycle.
- *Business insight:* Applying different stocking and promotional strategies to each product cluster is ideal.

5 Critical Assessment

5.1 Validity

- **Apriori:** The min_support threshold was swept across 7 values, with a visible elbow in the support curve. Rules are additionally filtered by confidence and lift, reducing the likelihood of spurious associations.
- **K-Means:** The convergence of multiple independent indicators — low silhouette scores across all tested values of k , the absence of a clear elbow, and the tight FM correlation visible in the raw data — provides validation for the null result.
- **DTW Clustering:** The low silhouette score (0.11) shows that cluster boundaries are somewhat fuzzy. However, the clusters are clearly separable to a fair degree on the STL comparison charts.
- **Product Categories:** The sample keyword-regex category assignment excluded approximately 49% of the dataset. If excluded products carry different seasonal profiles, the sampled clustering results may not represent the full product catalog. For best results experimentation with a wide variety

of different product categories and more in depth analysis of category clustering would be required.

5.2 Limitations

- **Geographic concentration:** 91.5% of transactions are UK-based. Findings reflect UK consumer behavior specifically and may not transfer to other markets.
- **Single-year window:** Seasonal patterns are confirmed over one year only. It is not possible to distinguish true recurring seasonality from a one-off annual anomaly without additional years of data.
- **RFM scope:** The null result reflects the limits of this particular dataset. A longer time window, a broader geographic mix, or the inclusion of product-category preferences as additional features may surface behavioral segments not visible here.
- **Category coverage:** Keyword matching is somewhat brittle. Products described with non-standard phrasing will be missed, and the sample of 15 categories cannot provide a complete picture of the product catalog.

5.3 Ethical Considerations

- **Privacy:** The dataset contains no personally identifiable information. CustomerID is an anonymous integer. Furthermore, all analysis is performed at the aggregate or segment level. Individual privacy risk is very low.
- **Manipulation:** Association rules could be used to construct upsell flows that exploit the observed collecting behavior. Responsible deployment should ensure recommendations are presented transparently, without manufactured urgency..

6 Conclusion

6.1 Discovery Questions Revisited

- (1) **What products are frequently purchased together?**
Answered. Strong association rules (lift up to 24.2) confirm systematic variant set purchasing.
- (2) **What is the distribution of spending across the customer base?**
Answered — Null result. RFM clustering found no meaningful segmentation structure. The customer base is homogeneous in behavior; customers differ in how much they spend, not in how they spend.
- (3) **Which product categories share similar seasonal sales patterns?**
Answered. Three seasonal categories were identified (holiday, winter-spring, no seasonality), but the DTW silhouette score of 0.11 indicates fuzzy boundaries. The clusters should be treated as guides.

6.2 Summary of Business Insights

- Bundle promotions and cross-sell recommendations should be organized around product variant families, consistent with the strong co-purchase rules identified.
- Spend-based prioritization is the appropriate targeting strategy on account of the segmentation null result.

- Inventory and promotional planning should be adapted to seasonal clusters.

6.3 Possible Future Work

- Investigate a broader selection of product categories for STL + DTW clustering.
- Analyze a wider set of data across several years if it exists.
- Perform customer segmentation with a much wider feature set. Incorporating product-category preferences or return rates could reveal customer segments that spend behavior alone does not show.